# CubeSat Launch Initiative
# SilverSat Final Report

## Project Description

SilverSat Limited is a multi-year, non-profit STEM organization for the Silver Spring community. SilverSat was originally started by three families in the Montgomery County Public School District in 2016. SilverSat Limited has developed into a student-focused aerospace program, giving students from grades 6-12 experience with leadership through designing, building, testing, and operating a CubeSat. We chose to make a CubeSat because it embodied the mission of SilverSat Limited, which is to provide a challenging and enriching STEM program for children in the Maryland and D.C area.

For our satellite we chose to demonstrate a mission sending data to its users directly through social media. Our CubeSat took photos of Earth from low Earth orbit and posted the photos to X (formerly Twitter), logging directly into X from the CubeSat. In the process we fulfilled the objective of teaching ourselves and others more about satellite technology. Our proposal was accepted by NASA in the spring of 2021. We developed a viable design, built and then tested our CubeSat over the next 4 years. SilverSat launched on September 14, 2025 and was deployed to orbit on December 2, 2025.

We successfully met our mission objectives. We received our first communications at our ground station 3 hours after deployment. For the next five months, SilverSat continued to operate and returned images of Earth which were either posted as tweets via our ground station or broadcast using Slow Scan Digital Video (SSDV) mode. We demonstrated in space the use of the Improved Layer 2 Protocol (IL2P), thus contributing to advancing the radio art. The number of tweets and images returned are given in subsequent sections. SilverSat ceased operations on April 23, 2026 when it reentered the atmosphere.

The subsystems we designed to accomplish this mission are shown in Figure 1. The 1U CubeSat components were housed in an EnduroSat open frame. The power subsystem included solar panels, batteries and control electronics from EnduroSat. The CubeSat antenna was also from Endurosat. Our Avionics board contained the main microprocessor, and provided timing, activity scheduling, collection of status telemetry and processing of uplink commanding. The Radio board handled all communications with the ground and operated at 9.6 kbps in the UHF amateur radio band. The Payload board hosted a Raspberry Pi Zero paired with a 4D Systems uCAM-III camera. The Raspberry Pi Zero stored the images, implemented the post function to X, and processed the images for the SSDV downlink. The Payload board combined with the radio implemented the IP-over-IL2P and SSDV-over-IL2P protocols. The Payload board, Radio board, and Avionics board were designed by their respective SilverSat teams, and manufactured locally. The assembled CubeSat containing all of these subsystem components is shown in Figure 2.

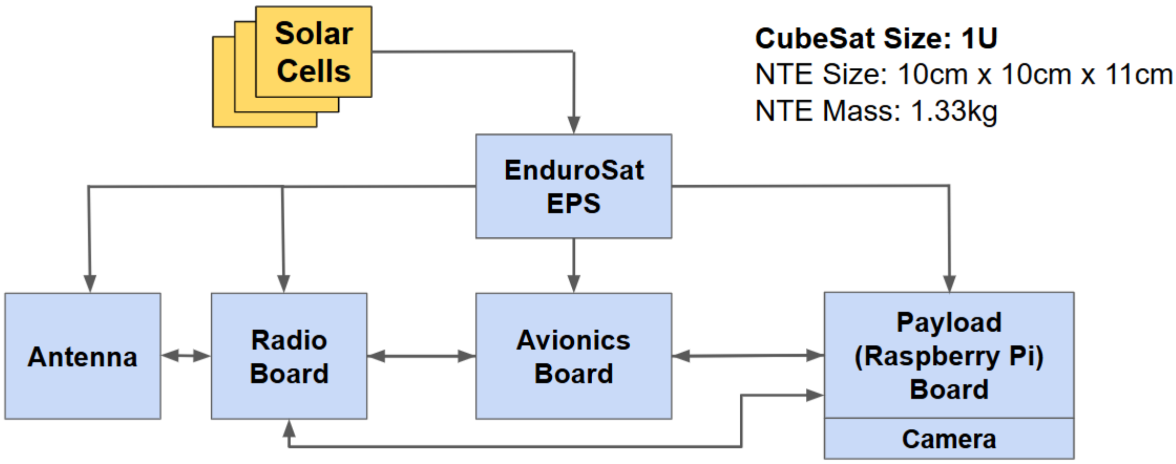



Figure 1. SilverSat Block Diagram

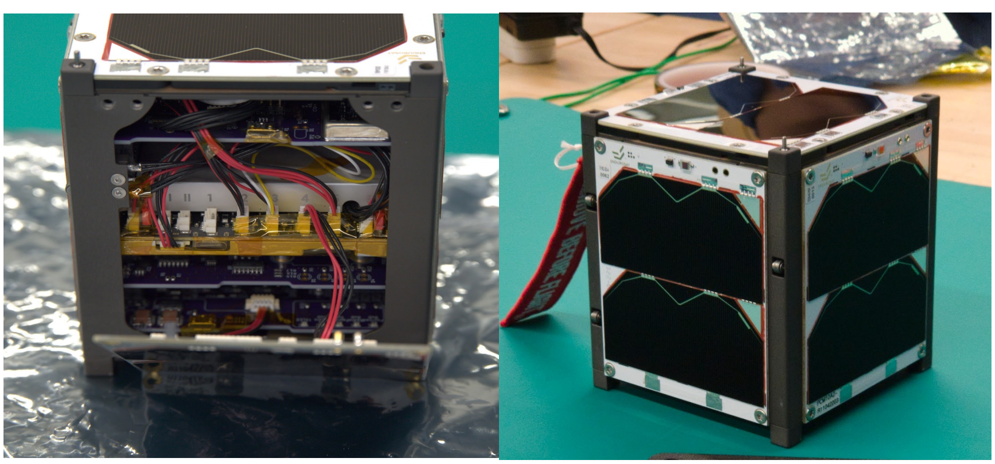
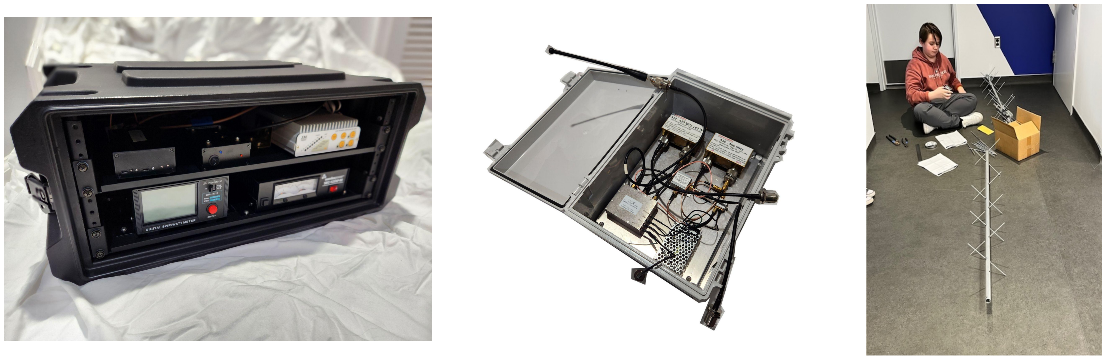


Figure 2. SilverSat CubeSat a) in construction b) ready for flight

We also built a ground station to transmit commands and receive data from our CubeSat, shown in Figure 3. The ground station radio is a copy of the CubeSat radio, to which external amplifiers were added: a power amplifier for the uplink and a low-noise amplifier (LNA) for the downlink. The ground station included a user interface to issue commands and receive responses and a facility to provide frequency adjustments to the ground station radio. The ground station allowed us to command the CubeSat, monitor its health, and relay the CubeSat-generated tweets to the Internet. Students earned their amateur radio licenses to operate the ground station equipment. Figure 4 shows members of our team conducting a pass using our ground station.



(a) Indoor ground station (b) Outdoor ground station (c ) Antenna

Figure 3. SilverSat Ground Station Equipment

Figure 4. Operating the SilverSat Ground Station

We were successful in demonstrating our concept of operations. The CubeSat consistently transmitted Morse code beacons and telemetry packets. Our operations team planned and uplinked a schedule for taking and transmitting images of Earth on a weekly basis. When the satellite passed over the SilverSat ground station, we uplinked this schedule and received telemetry on its health and status. We were also able to give the CubeSat access to the internet which allowed it to log into X and post the images as tweets. Between contacts with the SilverSat ground station, the CubeSat transmitted a picture using its SSDV mode, at set times that were included in the uplinked schedule. Global tracking communities followed SilverSat's beacon. Radio amateurs and enthusiasts around the world successfully received and decoded our pictures from the CubeSat that were broadcast in SSDV mode.

SilverSat opened up a dedicated "Operations" portal on our official website (silversat.org). Anyone from the public, schools, or the amateur radio community could log on and submit a request for the satellite to take a picture of a specific geographic coordinate on Earth.

As part of designing our CubeSat we developed targeted courses in different aspects of engineering techniques used in the design and in amateur radio usage. We developed documentation and technical specifications which have been made publicly available as a reference for other CubeSat teams. Our designs, hardware, and software are published on github at https://github.com/silver-sat. Students and mentors presented the project to the community at numerous events including many science fairs for elementary, middle and high school students. Our presentations included a description of our mission and CubeSat and samples of our hardware.

# Educational Accomplishments

The main purpose of SilverSat as a group is to educate students and the community about space technology. Within the group, students worked along with adult mentors to learn and to build the CubeSat, with all students encouraged to participate in the construction and testing of the CubeSat. This allowed our students to learn new skills, including electronics design, mechanical design, programming, project management, team leadership, and presentation skills. Through the process of programming the CubeSat and creating test scripts, many of our students learned how to code with programming languages such as Python, bash, Arduino C, and C++.

Additionally, in order to educate student members, SilverSat gave various courses to its students. One such course was on KiCad, which taught students about schematic capture and PCB design. This skill allowed our students to design the PCBs used on the CubeSat itself. In order to prepare for making radio contact with the CubeSat, SilverSat hosted amateur radio courses for its students. This taught many student members radio concepts and basic electronics. Six students and two adult mentors got their amateur radio licenses during their time at SilverSat. Several instructional presentations were developed and presented on related topics, including “How to Measure a Magnet,” “Orbits 101,” and “Solar Cells.” Through SilverSat’s relationship with Ansys, one middle school student even became proficient at STK (see https://www.agi.com/blog/2021/08/silversat-cubesat-launch-initiative)

Outside of SilverSat, we presented at many high schools, Science Centers, and technical conferences as shown in Figures 5 and 6. As a group, we did community outreach at various science centers and high schools in order to inform people about SilverSat’s activities and progress:

- An overview of SIlverSat was presented to a class at Capitol Technology University

(Nov 2025).

- We had a booth at Woodland STEM Night at Woodland Middle School in Silver Spring, MD (Nov 2025).
- We presented an overview of SilverSat to an engineering class at Wheaton High School (Sept 2025)
- We had a table at the Goddard NASA Visitor's Center 50th anniversary (May 2026)
- We had tables for Rockville Science Center's Rockville Science Day (April 2025 and April 2026).
- We had a booth at Family Science night at Howard B. Owen Science Center (Dec 2025).
- We had a booth at Maryland Science Center's Invent The Future Day (Nov 2025). Maryland Science Center also aided us in public outreach by publicizing our visit on their facebook page.
- We presented to a physics class on how to measure a magnet at Ramapo High School in New Jersey. (May 2025)

Our presentations included a description of our mission and CubeSat and examples of our hardware that others could interact with. This allowed hundreds of people to learn about the project, space technology, and engineering fundamentals.


Figure 5. SilverSat presentations at Howard B. Owen Science Center's Family Science Night

Figure 6. SilverSat presentations at Rockville Science Center's Rockville Science Day

# Technological Accomplishments

We succeeded in our primary mission goal of posting images to X from orbit. Our CubeSat took pictures of earth and posted them directly to X during contact with the SilverSat ground station. When in contact with the ground station the CubeSat set up as an IP node and formed a HTTP POST request to twitter's API. We had a total of 21 tweets from space with 7 being good pictures of Earth. An example post appears in  Figure 7. The other tweeted pictures of Earth appear in the Anecdotal information section.

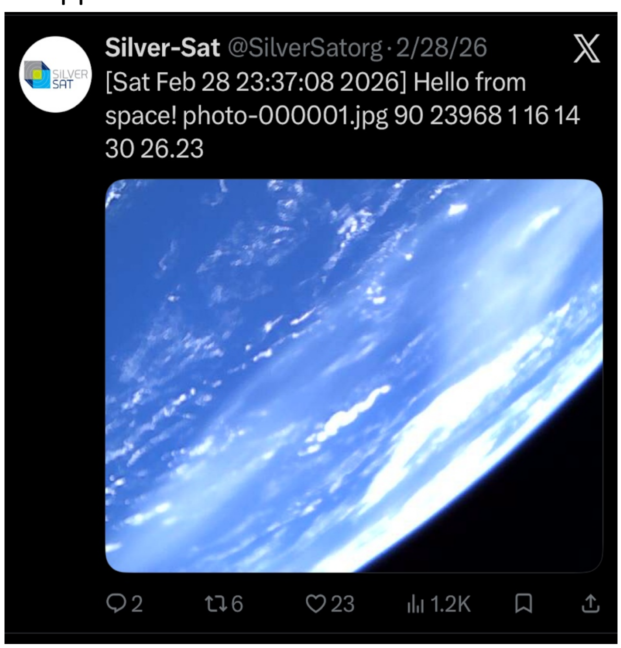


Figure 7. Tweet posted by the CubeSat

We demonstrated the use of IL2P and the combined stack of IL2P and SSDV in space.  IL2P is an emerging link layer protocol in the amateur radio community that provides more robust communications than its predecessor, AX.25. IL2P supports Reed-Solomon forward error correction (FEC), which makes it more robust for space use. SilverSat is one of the first CubeSats to implement this protocol. The complete payload communications model appears in Figure 8.

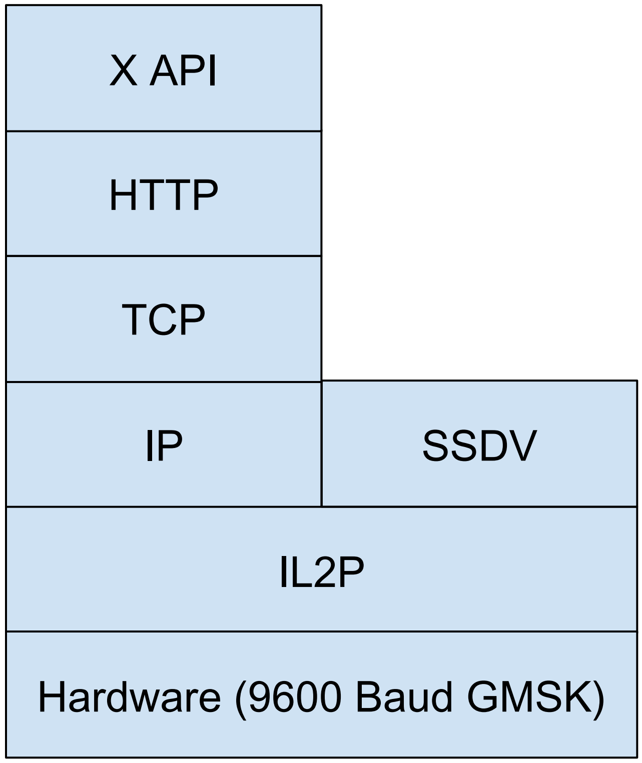



Figure 8. An OSI model of the payload downlink

SilverSat also used the open source Raspberry PI computer as the payload processor; this helped validate this inexpensive, user friendly, and widely used computing system for space flight. This opens up a whole computing ecosystem that many grew up using as an effective validated way of computing in space. The Raspberry Pi controlled the camera, handled the exchange with X, and implemented both the Internet protocols and SSDV.

Another technical investigation that was needed during development involved the magnetic fields of the magnets used for attitude control and rods that were part of the structure. We noticed that the magnets could stick to the M304 stainless steel rods that run along the edges of the CubeSat. This led to worry that the rods would absorb some of the field, reducing their effect and degrading our attitude control capability. As shown in Figure 9, we ran an experimental test to measure the effect that the M304 stainless steel rods had on the strength of the magnets when compared to brass rods (cut to match moment of inertia) by comparing their resonant periods when floating in water. We found that the stainless steel rods caused a net 3.8% decrease in the effective magnetic moment, which is notable but did not affect our CubeSat design.

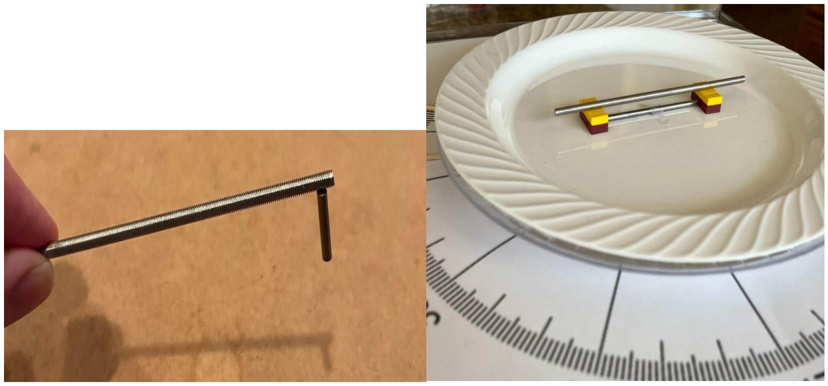


Figure 9. Experimental setup for measuring effect on effective magnetic strength by stainless steel rods

# Scientific Accomplishments

SilverSat was a technology demonstration mission and thus was not meant to gather any scientific data. However, SilverSat demonstrated a potential way to transmit time-sensitive data from scientific missions. We demonstrated posting the images SilverSat took to X directly from the satellite through a ground-hosted Internet connection. This capability could be used to send data directly to a mission's user community so that they know when something of note has occurred. For example, this system could allow automatic alerting of space events such as solar flares and gamma ray bursts. The satellite could post a message to social media whenever it detects an event, and the public and various scientific communities will be able to receive this information by enabling notifications on the satellite's social media page. This eliminates the need for the mission to first receive the data and then separately send it out to users, thus reducing the delay for time-critical data. This capability would be especially useful if combined with a relay-based communications system for continuous contact coverage.

# Impact

SilverSat Limited's impact from our first CubeSat project was in four areas:

- Our students
- The amateur radio community
- The general public
- Future CubeSat missions

## Our Students

SilverSat Limited has mentored 45 students from 34 schools in the Washington, DC metropolitan area. Thirty-four of our students started in middle school and stayed with the program through high school. Five of these 34 stayed in SilverSat through high-school graduation. Among 18 past or present students surveyed:

- 15 either are currently studying or have graduated with STEM degrees from university.
- Five SilverSat graduates are either working or interning in technical fields.
- Eight students and mentors earned amateur radio licenses.

## The Amateur Radio Community

As described in the technical accomplishments section, SilverSat provided an accessible platform for the amateur radio community to receive our beacons (Morse code) and SSDV signals (images). Since deployment, 46 certificates of successful signal reception (QSL cards) have been requested by amateur radio operators and satellite enthusiasts on all continents except Antarctica (Figure 10). Our mission has also now demonstrated the use of IL2P in space. Many members of the community received our SSDV signals during numerous opportunities during operations as shown in Figure 11.


Figure 10. SilverSat QSL card

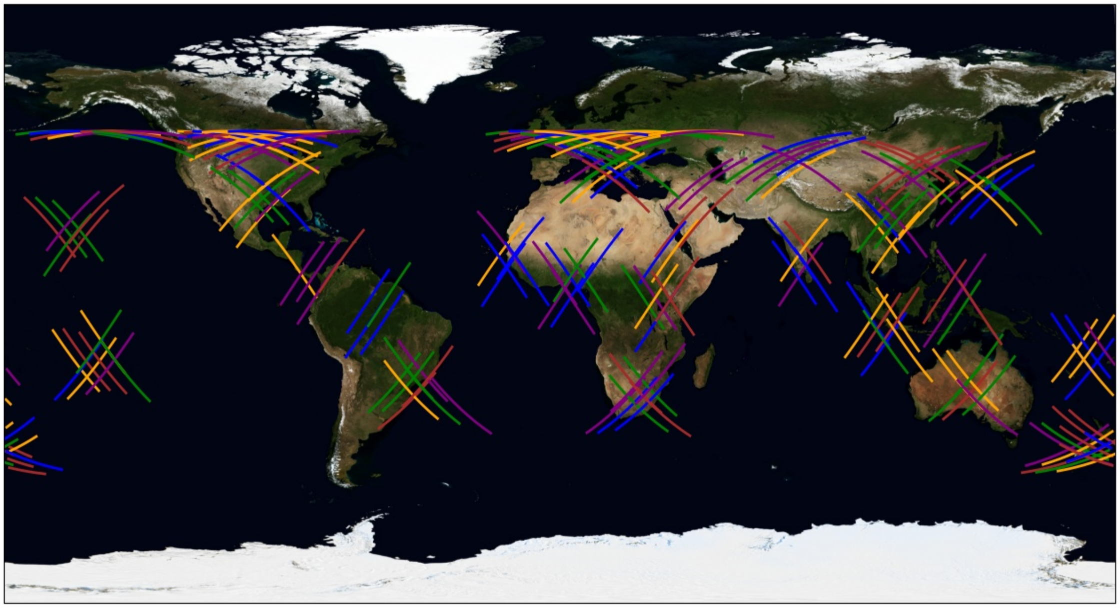


Figure 11. SSDV passes for March 30 through April 6, 2026. The colored lines are ground tracks when SSDV was active.

Hobbyists interacted with other enthusiasts and SilverSat Limited using their own independent forums (https://community.libre.space/search?q=SilverSat). One community member used SilverSat's SSDV transmissions to develop their own free and open source decoder (https://github.com/hobisatelit/ssdv2sat).

## The General Public

SilverSat Limited's public outreach included presentations at community events(described in Educational Accomplishments), postings to social media, and news appearances. As of June 13, 2026, our X page (https://x.com/SilverSatOrg) had 331 followers. Local media covered SilverSat's launch (http://www.fox5dc.com/video/1707573 and (https://www.mymcmedia.org/student-created-satellite-launched-into-space/) and the satellite was featured on YouTube (https://www.youtube.com/watch?v=qvN3sE2Nv4U).

## Future CubeSat Missions

We believe our designs and our implementation have the potential for reuse given the success of our mission. All of our hardware designs, software, and associated documentation are freely accessible on GitHub (https://github.com/silver-sat/). The amateur radio community's open source software was later expanded to support other missions such as SpinnyONE (HADES-SA). We will also use this as a resource for our future CubeSat missions.

Additionally, SilverSat members currently work with several other organizations helping them start their own CubeSat missions.

# Development Challenges

Throughout the development of the CubeSat many challenges were addressed in 4 main areas: Technological, Personnel, Logistical, and COVID-19.

## Technological Challenges

The first design of the payload board had a flaw due to the lack of Universal Asynchronous Receive-Transmit (UART) lines on the Raspberry Pi, forcing us to use the Serial Peripheral Interface (SPI) pins and an external UART to communicate with the camera while radio took up the serial lines. This caused significant problems when attempting to transmit an image from the camera to the Pi, such as long transmission times and frequent timeouts during transmission, preventing the picture from uploading to the Pi. The result was that the Pi needlessly expended power while attempting to transfer an image and sometimes would not complete the transfer. The eventual solution to this problem was to implement a UART switch, as the Pi should theoretically never have to communicate with both the camera and the radio simultaneously. This resulted in a much more effective method of transmitting the images and completely resolved the issue.

The International Amateur Radio Union (IARU) coordinates frequencies among satellite missions planning to use  the amateur radio satellite bands. Their objectives include promoting technical and scientific investigations in the field of radiocommunication and enhancing amateur radio as a means of technical self-training for young people. SilverSat engaged with the IARU late in the development phase. On our first application, the IARU expressed that the mission did not match requirements for use of amateur frequencies.  After consulting with a member of the IARU, we added the SSDV mode and the IL2P demonstration to better match these requirements. Coming late in the development phase, we needed to rush their development.

Another problem we encountered was the balancing of the power budget, which caused significant shifts in our design. In the original design, it was estimated that the battery would fully drain if the Pi and radio were on for prolonged periods of time, which would cause the satellite to become dormant and reboot once it recharged. This ended up not being an issue as the battery worked far better than expected but at the time we treated this as a genuine concern. To combat this issue, we implemented a change in avionics which would remove power to the payload board, and thus the Pi, if payload remained active for more than ten minutes. This did not end up occurring during the mission, due in part to various changes to the payload board such as the aforementioned addition of a serial switch which reduced the time that the payload board remained active.

During the assembly of the CubeSat's Flight Model, we accidentally installed the stabilization magnets in the opposite direction, which would have caused the camera to point away from the northern hemisphere. A student developed a test plan which caught the issue before the CubeSat was launched. We resolved this issue by installing the magnets used in the Engineering Model.

Our original design included majority voting on three digital input/output (I/O) lines to enable the avionics board to signal the payload whether to take a photo or transmit a tweet. The addition of an SSDV mode required the introduction of a third alternative. The hardware implementation was complete when this requirement was added. We implemented an extended solution based on precise timing of the digital I/O signals.

The potential for I2C bus lockup was identified during system testing. We later learned that this is a very common cause of CubeSat failures. Our design included both a critical and non-critical I2C bus, mitigating the risk to some extent. We included software to identify a bus problem and to unlock the bus in the event of a loss of synchronization.

The ground station software needed to manage multiple asynchronous data streams: user input, data from the radio, commands to the radio, and doppler adjustments from Gpredict. We also needed a flexible user interface. We solved these problems using a local sqlite3 database for serialization, interfaces to the database written in Python, and the use of Flask to provide the user interface in a browser. The design proved to be flexible and extensible.

The Arduino process loop implementation was a limitation for our avionics system. Since all active code needs to be accessible within the loop, timing and reliability issues in any subsystem can easily cause a system-wide impact. The challenges were compounded with the changing requirements for various delay intervals and for access to functions during initialization and self testing. We addressed the challenges with a consistent approach to implementation of both the initialization software and the loop software, the use of software state machines for selected critical tasks spanning multiple loops, and object-oriented techniques to manage complexity and control scope.

## Personnel Challenges

SilverSat met as a club on Saturdays, with virtual side meetings during the week. For both our students and mentors, their ability to participate varied often unpredictably over time due to other commitments such as jobs, clubs and sports. This made planning difficult for key activities and deliverables that were dependent on specific students. We also struggled as ultimately students left the program either for college or to pursue other interests.  We ran into issues with different systems because the people who had designed said systems had left the program.  We had several instances where senior students diligently trained their successors; in other cases we had to quickly train junior students to take their place. We also had difficulty finding mentors whose skillsets collectively spanned all the needed topics.

## Logistical Challenges

Due to the small and mostly decentralized nature of the project, we never had a proper meeting space controlled by SilverSat, and operated mostly by finding spaces that would allow us to work in them. Because of this, we never had a space in which we could store our materials long-term. This meant that we had to transport everything we needed for our meetings back and forth between our homes and our meeting place. Additionally, certain spaces had restrictions that would make it difficult to do certain tasks, such as soldering, within those spaces.

## COVID-19 Pandemic Challenges

During the pandemic, we were unable to work on any physical aspects of the CubeSat as a full group outside of exceedingly rare circumstances. This significantly slowed our progress and hampered the actual construction of our prototype. Some circuit prototype work was done by students at their homes during the pandemic, with kits of parts dropped off at their doorstep.

Various important presentations and documents were created asynchronously and virtually. This included holding our Feasibility Review, writing the mission proposal, and holding our Preliminary Design Review after award. The separation generally made it more difficult to come up with one coherent document in an efficient manner, as well as forcing us to present virtually.

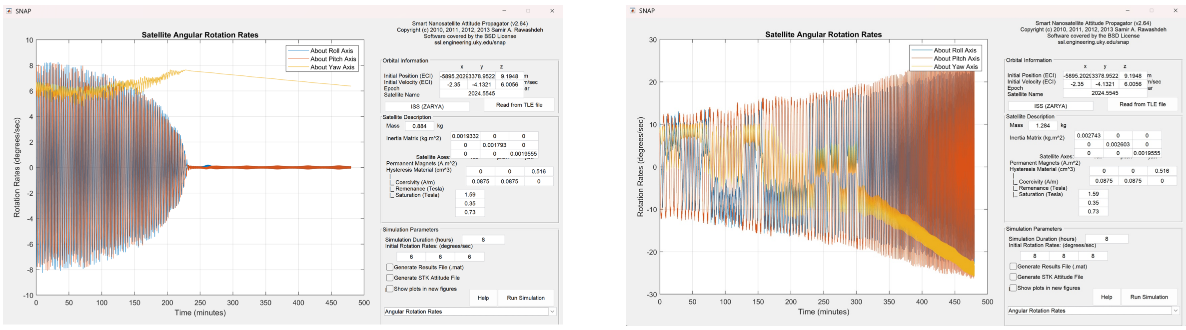
Retaining interest from students was even more challenging throughout the pandemic than before or after it. This was due to the fact that students and mentors were forced to work together completely virtually, as well as being unable to do hands-on work on the satellite design.

# Mission Challenges

During the Mission, challenges were encountered but that did not stop us from meeting the mission objectives.

## CubeSat Stabilization

Our passive magnetic stabilization system involved magnets and hysteresis material, based on the design used by University of Kentucky's KySat. Simulations were run which brought us to the conclusion that the CubeSat would stop tumbling within about 2 hours after launch. However, we did not properly account for how much the CubeSat would be spinning in the initial conditions of our simulations. Thus, the CubeSat ended up tumbling and did not properly stabilize while in orbit. Figure 12 shows results of updated simulations run after launch showing the existence of a threshold in initial spin rate which forms a border between proper settling and chaotic tumbling, somewhere between 6-8 degrees per second. The error in thinking during design was that a higher initial spin rate would simply cause a proportionally longer time of settling, which would have been fine.




(a) Detumbling simulation with initial condition of 6 degrees per second per axis, where the CubeSat detumbles properly.

(b) Detumbling simulation with initial condition of 8 degrees per second per axis, demonstrating that the magnetic stabilization system fails at higher initial conditions.

Figure 12. Attitude Control Simulations of SilverSat

Because the magnetic stabilization system did not stabilize the CubeSat's attitude, we did not have confidence in which direction the satellite was pointing when we wanted to take pictures. We applied for and received a NOAA imaging license modification that recognized the CubeSat would take photos with uncertainty of the camera direction. While the CubeSat took a number of pictures of Earth, a significant number of our images were taken with the camera pointed into space. From this, we learned that magnetic stabilization requires careful design and is very difficult to test on Earth. One point now understood from the simulations run after launch is that it is difficult to distribute the mass in a 1U CubeSat to avoid issues associated with the intermediate axis theorem. The KYSAT reference paper we based the design upon was a 3U cubesat which naturally has a preferred axis of spin, and is thus easier to stabilize. We should consider other stabilization alternatives for future 1U missions unless we are able to substantially redistribute the mass.

## CubeSat Communications Problems

After deployment and activation, the CubeSat's beacon indicated that the antenna had not deployed properly. The beacon character that indicated the antenna fault, by design, would be frozen at this value until the CubeSat reset. The CubeSat reset twice throughout the mission, which allowed it to retry deploying and checking the antennas. The CubeSat continued to indicate the fault, meaning that the antenna never deployed properly. This fault, in combination with the CubeSat stabilization issues, affected the

CubeSat's antenna gain pattern and caused large variations in the radiated power as the CubeSat spun. We were able to continue with the mission with adjustments to the ground station to improve its sensitivity and reduce packet loss from the CubeSat. Figure 13 shows the received power for a typical pass. As shown, the received power could vary as much as 7 dB. Because of this challenge, we learned that it would be useful to have more detailed telemetry from the antenna to better clarify which errors the CubeSat is facing, and it might be useful to have a command to retry antenna deployment.

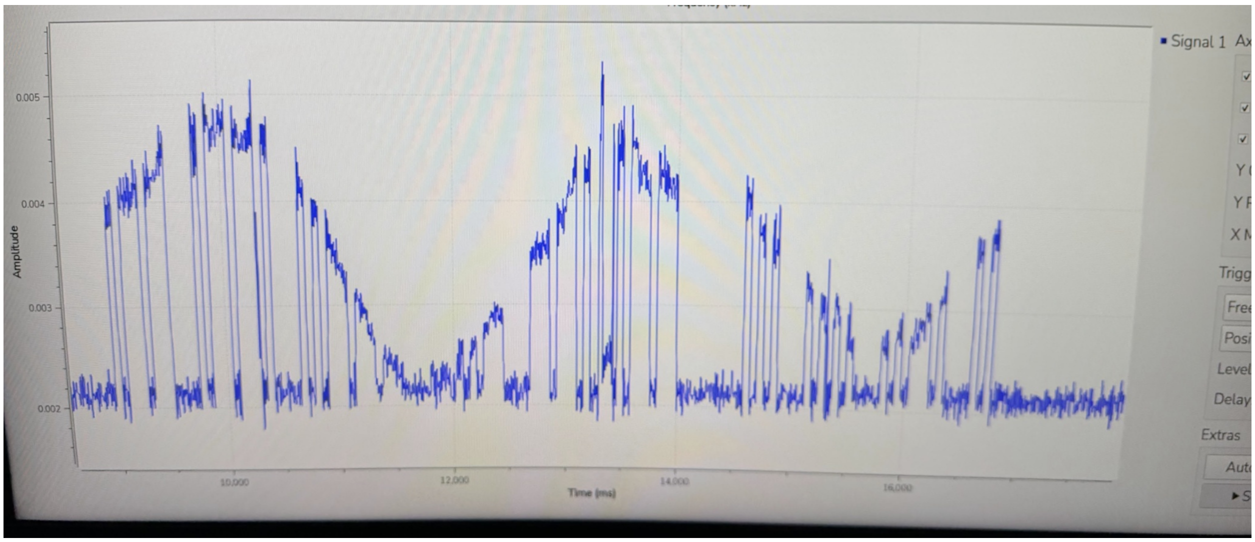



Figure 13. Graph demonstrating the power received at the ground station over time. The gaps in the curve indicate the packetized nature of the data. The overall curve is a combination of the partially deployed antenna and the CubeSat tumbling.

The CubeSat also struggled with communications due to a bug that went undetected in the radio board's software that would occasionally turn off the PA (Power Amplifier) during a packet transmission. This erratic deactivation would result in packets being transmitted with significantly less power than expected. This bug was not detected due to our radio communications testing not being at flight-like power levels resulting in us not noticing the PA dropping out.

## Ground Station Modifications

We needed to improve the sensitivity of the ground station to overcome the CubeSat antenna deployment fault. Further, pre-launch ground testing did not reveal some problems at the ground station. Because of these issues, a number of changes were made at the ground station right after deployment. The ground antenna was not pointing correctly so it was calibrated so it pointed more accurately. We implemented a sequencing system to use the available mast-mounted low-noise amplified (LNA) in addition to the LNA in the ground station, improving the ground station's noise performance. We were expecting the CubeSat antenna to be Left Hand Circularly Polarized (LHCP) but we quickly discovered we got better reception if we switched to Right Hand Circular Polarization

(RHCP).  We had implemented the ability at the ground station to switch polarizations,  Our ability to debug the antenna issue on the spacecraft was limited by the telemetry we had made available to the ground. In future missions, we would expand the onboard telemetry that the ground could access.  We also recommend carrying a higher link margin on CubeSat links when using new designs to handle unforeseen issues.

## Ground Station Operations

The SilverSat ground station was located at the amateur radio club at the Goddard Space Flight Center (GSFC), which was over 40 minutes away from most of our members. This, along with many CubeSat passes occurring at odd hours made scheduling difficult. Site access for wintertime operations was difficult during times when significant snowfall had occurred. As a result, it was often hard to schedule times for members or mentors to operate the ground station. Another obstacle was the requirement of at least one member present with an Amateur Radio license and a member of the GFSC club, further complicating our efforts to coordinate. Through this, we learned that managing CubeSat operation at a fixed ground station can be very difficult on a changing schedule. It is important to plan detailed roles and procedures for each person interacting with the satellite. Additionally, having remote access would be very beneficial.

## Tweeting

Tweeting was challenging due to a number of issues. The system was designed to tweet the oldest photo taken that had not been tweeted, a prioritization that was not always desired. We wanted to skip photos that were not clear and those that did not show the Earth. The payload software included a function that would retrieve a small command script from the ground station just before posting a tweet. We adapted this function to re-order photos on the Payload and control which photo was posted.  Using this method, we were able to achieve the objective of tweeting a photo of Earth from space. This taught us to provide easier tools for updating flight software during the mission and to build flexibility into the payload scheduling process.

After contact had been established with the satellite, we attempted to have the satellite connect to the Internet through the ground station and tweet. However, our initial attempts were unsuccessful due to aforementioned communications issues with the CubeSat. Challenges also arose with the TCP connection with X, where the communications issues above interacted with the protocol's retransmission.  Solutions could only be implemented on the ground station side as modifying the software on the payload board would have been extremely risky due to the unstable link. The solution to our problems was twofold:

1. A ground-side HTTP Proxy that stood in between the satellite and the X/Twitter endpoint was added. This required the use of NAT (Network Address Translation) that patched the satellite into the proxy instead of directly to the X/Twitter

endpoint. The proxy received the request from the satellite, parsed, cleaned, and injected the address of the proper endpoint into the request and sent it off to stunnel, a proxy we can send packets through to get encrypted, which managed communicating with the X/Twitter endpoint. Before stunnel could communicate with the endpoint, it needed to add TLS (Transport Layer Security), a method of encrypting and securing data being sent via the internet, as it is part of the strict requirements that X/Twitter requires in order to communicate with its API.

2. Modifications needed to be made to communication between the satellite and the ground station. The size of packets was reduced to avoid fragmentation, a process used in TCP/IP to send large packets through links that can't carry much information per transaction, as it led to fragments being lost and data being impossible to reassemble. TCP was also forced to not send or queue to receive a large amount of packets as otherwise the link could lose stability during a long stretch of sending or receiving packets.

After these fixes were implemented, we were able to successfully start tweeting via the satellite, with our first successful in space tweet being completed on Jan. 18th 2026.

# Lessons Learned

During the development of SilverSat we learned many things that we would have done differently if we tried this again. Students, mentors, and board members were able to write down these lessons to share them with the rest of the team. Compiled here are many of those lessons, suggestions, and some things we would do again in the future.

| **Title** | **Lesson Learned** | **What to do next time** |
|---|---|---|
| Memory Management | Images taken during ground testing were left in the CubeSat's memory at launch. These images had to be downlinked from orbit before images taken from space could be received. | Consider adding file management to allow more flexible access for a future CubeSat. Provide a capability to designate which images or files are to be transmitted. |
| Power Capacity | The Endurosat battery and power system worked well. The available power on orbit was higher than predicted: the battery remained at full charge for most of the mission. | The battery functioned as it should, if not better, and there is no need to restrict operations due to power concerns. |
| Communications Protocol | Our CubeSat used IL2P and AX5043 protocols to communicate with the ground station. | Consider using a more advanced communications protocol such as the CCSDS (Consultative Committee for |

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| | | Space Data Systems) protocol. Using and then publishing a more advanced stack such as the CCSDS stack would add capability for the amateur radio community. |
| Attitude Control Method | Satelite's passive magnetic stabilization had issues resulting in tumbling once in orbit. | Consider using other options for stabilization to ensure stabilization and camera facing towards Earth. Expand dynamic simulations to determine bounding capabilities of selected methods. |
| Power Management | The avionics board ran continuously but could have been powered down when not needed to extend battery life. | Consider enabling a low power mode for when the board must be on but isn't actively doing anything. |
| Protecting Command Transmission | Implementation of command authentication using a signing protocol between the ground station and the CubeSat ensured efficient commanding. However, it required attention to detail in the implementation and also required shared secret management for authentication of commands. | Continue to use command authentication for future CubeSats. Review implementation for possible simplifications in design or more convenient use in operations. |
| Custom Hardware | SilverSat designed three custom boards for the CubeSat. These boards were an invaluable learning experience for the students. | If you have the team for it, don't shy away from custom hardware design. Assess your team, both students and mentors. In areas where you have strengths, use custom hardware designs to teach the students engineering development in depth. |
| Software development | Standard software tools and services worked very well for our mission. In the late stages, we started to use AI-based tools. | Given the remarkable productivity of AI-based tools, we will use them extensively in future missions. |

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| Prototyping, and making design decisions to increase lesson opportunities. | SilverSat prototyped both circuits and code, which were excellent learning opportunities. Some functions, such as the real-time clock, were implemented as separate circuits so we could facilitate building a prototype as a lesson. | Use portions of the design as teaching opportunities to let students gain experience. For projects such as this, educational benefit is more important than an optimized design. |
| Project documentation | Good technical documentation was a significant advantage and led to a better implementation and fewer errors and regressions. One example was our documentation of the radio/avionics interface that allowed team members to easily work on both subsystems independently. | Good documentation should be prepared for all mission software and interfaces. |
| Software version control | Programs such as KiCad change versions and new versions may not be backward compatible with previous versions. | Select the set of software tools to be used at the start of design including designated versions. Then make clear decisions to upgrade to new versions, switch to different tools, or stick to the original version over project lifetime. |
| Test campaign | Testing wasn't as specialized as it may have needed to be. | Testing should be more than "works" or "doesn't work", Define test objectives in advance and find root cause(s) for why something does not work when tested. In addition, test individual parts separately from the whole, instead of waiting for all components to be ready for integration.<br>Consider automating tests and adding regression testing after making changes. Explore using simulators or emulators to improve test programs. |
| Flight-like testing | Certain errors in our radio | Testing needs to be |

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| | communication link were masked by testing being performed under conditions that did not match actual flight. | performed with as close to a flight-like environment as possible. |
| Documenting Activities during Ground Contacts | Visit logs worked very well to keep track of activities during satellite contacts. | Continue using logs for satellite contact visits and keeping them organized and easily accessible to team members. |
| Physical presence required at ground station | Few members on the team were members of Goddard Amateur Radio Club, where the ground station was located, which limited the ability to schedule passes with the CubeSat.<br>This station could not be accessed remotely/virtually. | Ensure many members have the ability to access ground stations when needed. Develop capability to remotely use ground stations to communicate with the satellite. |
| Operational software deployment | We used a laptop to run our ground station software. It worked well, but was limited with regard to geography and multiple access. | Move the ground station software functions to a cloud service provider to enable access from multiple locations. |
| Built-In Test (BIT) Telemetry | Additional Telemetry would have helped in debugging antenna deployment and attitude control issues more quickly. In particular, sampling each of the 4 antenna doors on request and having an onboard magnetic alignment sensor would have provided needed insight. | Be more thorough in onboard Built in Test (BIT) capability and sample over full orbit to download on request, rather than only activating BIT during a pass over the ground station. |
| Team roles after launch | Many students didn't have a role or know what to do after the launch of the satellite. | Define and assign roles for post-launch operations prior to launch. Give roles to team members whose pre-launch roles are not needed after launch. Ensure all team members have access to the schedule for when they are needed to work with the satellite. |
| Communication outside of meetings | Lack of plan/agenda for meetings. | Discussing during previous meetings an agenda for the |

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| | | next meeting or having plans set over emails during the week leading up to the next meeting. |
| Material storage | Lack of ability to store materials between meetings caused shuttling problems and missing items. | Find a meeting place where storage is included in the meeting area. |
| Engage the amateur radio community early | We added functionality to the mission late in the development to better align with the amateur radio community. | If you plan to use amateur radio frequencies, work with AMSAT and the IARU early in the mission development. |
| LIbre Space community services | Libre Space is a worldwide community of satellite enthusiasts; they were very helpful with our CubeSat operations. For example, they performed orbit determination and helped us receive images. | Continue to use Libre Space capabilities for future missions. They maintain a worldwide network of ground stations that could augment or replace the ground station used for this first mission. |
| Team-building activities | Lack of a connected group feeling. | Holding times outside of set meeting hours for team building and/or bonding activities in order to feel more like a team. |
| Names on satellite | Students liked having their name or signature on the satellite when it was launched. | Offer this option to team members for future launches. |
| Amateur radio license | Students enjoyed the ability to study and test for licenses through the SilverSat program. | Continue these classes and this option. |
| Team composition | Most members learned of the program through word of mouth. As a result we reached fewer local communities than we would have liked to. | Make conscious effort to distribute flyers and recruitment papers in other languages (e.g. Spanish) and in communities often underrepresented in STEM fields. |
| Delivery of satellite data to the public | Our conops included tweeting images using X but SilverSat's target audience was not on X | Identify most popular social media tools in initial design and provide flexibility to |

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| | when we launched. | change to other tools as audience needs or composition change during operations. |
| QSL Cards Request Form | No field in form for recipients to enter their address, addresses are different in other parts of the world. | Include a field in form for address of recipient and email address to get in touch if there is an issue. In addition, add a field to request electronic QSL cards in form. |

# Anecdotal information

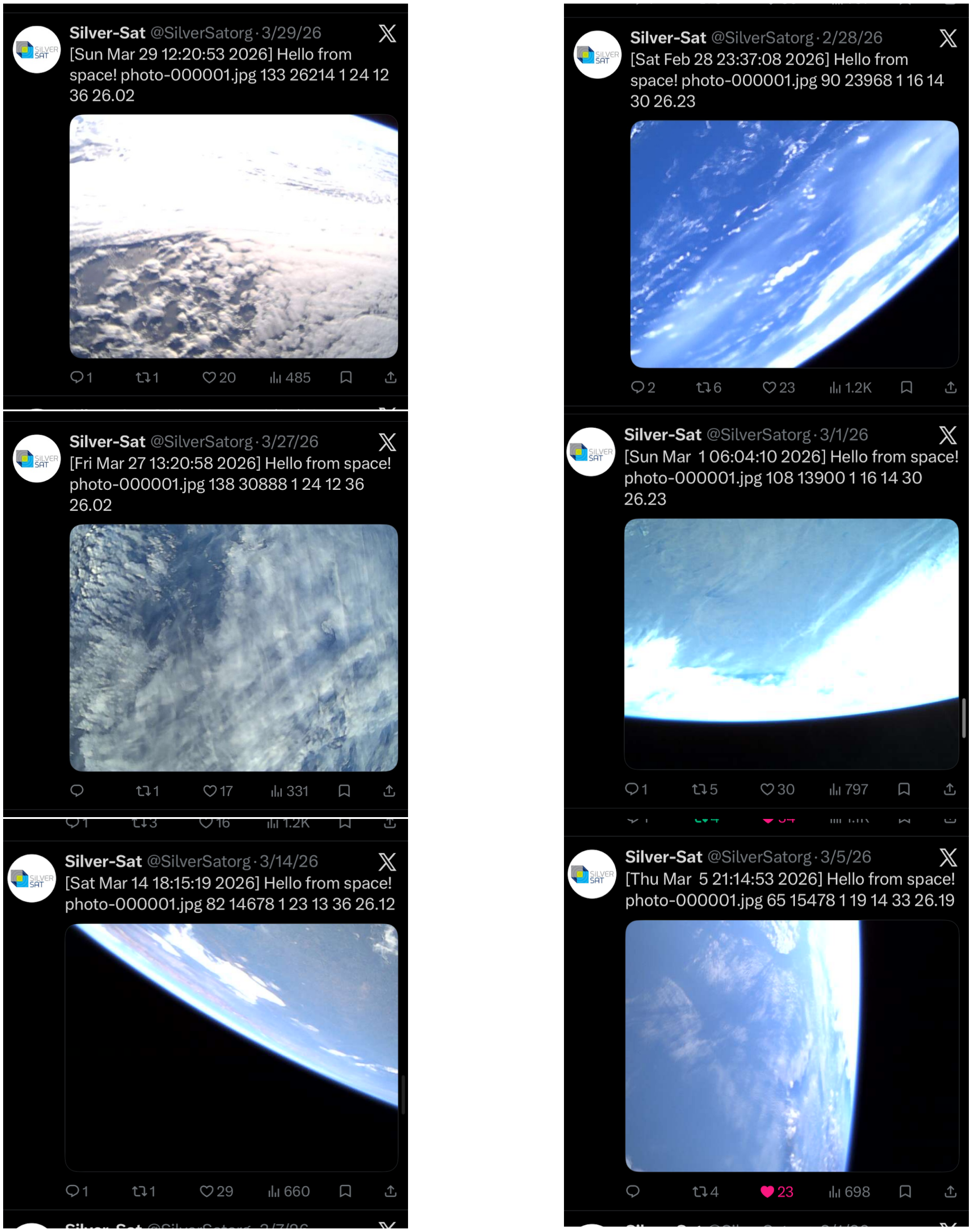


Figure 14: Gallery of SilverSat's pictures of Earth. The complete collection of all photos can be found on https://x.com/SilverSatOrg

# Publications

Webster, Daniel, “Design and Logistics Challenges Associated with Building a Nanosatellite - A Payload Team Leader Perspective,” AIAA Young Professionals, Students, and Educators (YPSE) Conference, JHU Applied Physics Laboratory, February 3-4, 2021.

Suko, Cade, “Passive Magnetic Attitude Control for SilverSat’s CubeSat,” AIAA Young Professionals, Students, and Educators (YPSE) Conference, JHU Applied Physics Laboratory, November 22, 2024.

Both of these authors were SilverSat students.